

Mag4D-SLAM Dataset: A Repeated-Traversal Multi-Modal 4D Geomagnetic Dataset for Localization and Mapping

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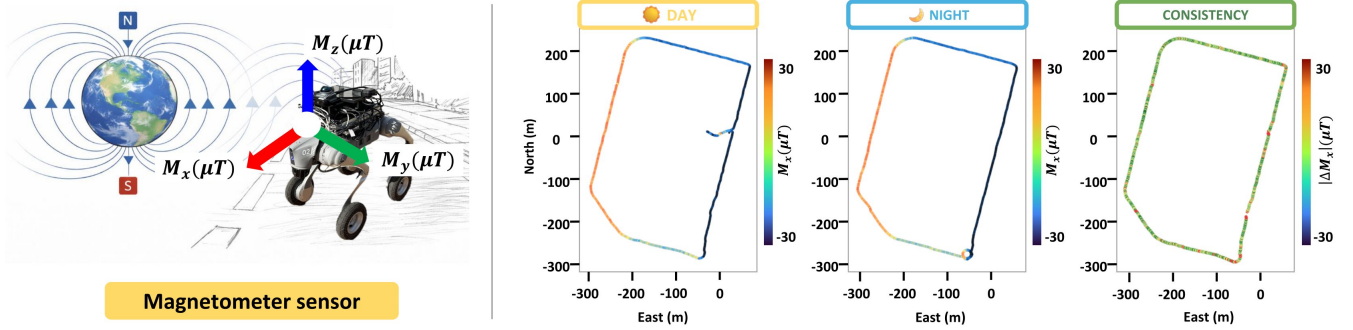


Fig. 1. Overview of the Mag4D-SLAM dataset and sensing paradigm. **Left:** The Earth’s geomagnetic field provides a globally available orientation reference measured by an onboard tri-axis magnetometer (M_x , M_y , M_z) mounted on a quadraped robot platform. **Right:** Spatial magnetic field maps (M_x) along repeated campus trajectories under (left) daytime and (center) nighttime conditions. The rightmost panel shows the cross-session consistency (daytime and nighttime) map ($|\Delta M_x|$), demonstrating that geomagnetic signatures remain spatially stable across day–night sessions, enabling illumination-invariant localization and heading estimation.

Abstract—Geomagnetic sensing offers an infrastructure-free, absolute orientation reference that is robust to GNSS denial and visual degradation, yet no large-scale outdoor robotics dataset supports its systematic study in SLAM. Existing magnetic datasets are confined to small-scale indoor environments and lack the synchronized multi-modal sensing, repeated-traversal structure, and high-precision 6-DoF ground truth required for geomagnetic SLAM research. We present Mag4D-SLAM, the first large-scale outdoor geomagnetic SLAM dataset. It comprises 14 sequences totaling over 18 km of synchronized LiDAR, camera, IMU, tri-axis magnetometer, and GNSS measurements with SE(3) ground-truth poses, collected along structured campus trajectories under paired day/night conditions in both forward and reverse directions. Through repeated-traversal experiments, we analyze three core properties: magnetic field repeatability across different recording sessions (daytime and nighttime), drift-free global heading estimation, and location-discriminative magnetic signatures for cross-session place recognition. Mag4D-SLAM is designed to support research on yaw drift mitigation, magnetic loop closure, and long-term localization — and to open new research questions on how geomagnetic sensing can complement visual and LiDAR modalities or provide a fallback cue under illumination changes, structural repetition, and GNSS-denied long-term operation.

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I. INTRODUCTION

Simultaneous Localization and Mapping (SLAM) is a fundamental capability for autonomous robotic systems operating in complex and large-scale environments. Over the past decade, significant advances have been achieved through LiDAR-based SLAM [1], visual–inertial odometry, and tightly coupled multi-sensor fusion frameworks. Public benchmarks such as KITTI [2], EuRoC [3], TUM-VI [4], Oxford RobotCar [5], and Boreas [6] have played a crucial role in accelerating research by providing synchronized multi-modal data with accurate ground-truth poses. However, these datasets primarily emphasize vision, LiDAR, radar, and inertial sensing, without incorporating geomagnetic measurements as a core modality.

In contrast, the geomagnetic field, despite being globally available, infrastructure-free, and inherently absolute in orientation, remains largely absent from robotics-oriented SLAM datasets. Existing magnetic datasets such as UJIIndoorLoc-Mag [10], MagPIE [7], and MagWi [9] are confined to indoor environments and lack synchronized visual–LiDAR–IMU–magnetometer measurements with high-precision 6-DoF ground truth. More recent work such as Mag-Match [11] addresses magnetic map registration but does not provide a large-scale outdoor dataset for evaluating magnetic-aware SLAM.

Beyond datasets, several works have incorporated magnetic information into SLAM frameworks. Osman *et al.* [12] modeled local magnetic anomalies using Gaussian processes within a Rao–Blackwellized particle filter for pedestrian SLAM. Wang *et al.* [13] proposed an orientation-aware 3D SLAM system using alternating magnetic fields from

TABLE I. Comparison of Public SLAM and Magnetic Datasets

Year	Dataset	Seq.	Avg. Length	Modalities	Mag.	Environment	Platform	Repeated Trav.	Day/Night	GT Pose
2013	KITTI [2]	22	~2 km	L, C, I, G	×	Outdoor	Car	×	×	✓
2016	EuRoC [3]	11	~80 m	C, I	×	Indoor	MAV	×	×	✓
2017	RobotCar [5]	100+	~10 km	L, C, I, G	×	Outdoor	Car	✓	✓	✓
2017	MagPIE [7]	10	~400 m	M, I	✓	Indoor	Handheld	✓	×	✓
2018	TUM VI [4]	28	~200 m	C, I	×	Indoor/Outdoor	Handheld	×	×	✓
2020	Newer College [8]	4	~1.5 km	L, C, I	×	Outdoor	Handheld	×	×	✓
2021	MagWi [9]	100+	Indoor multi-floor	M, W, I	✓	Indoor	Smartphone	✓	×	×
2023	Boreas [6]	40+	~8 km	L, C, R, I, G	×	Outdoor	Car	✓	✓	✓
2026	Mag4D (Ours)	14	~1.3 km	L, C, I, M, G	✓	Outdoor	Quadruped	✓	✓	✓

Abbreviations: L = LiDAR, C = Camera, I = IMU, M = Magnetometer, G = GNSS, R = Radar, W = Wi-Fi.

powerlines. While these works demonstrate the feasibility of magnetic mapping, they rely on task-specific experimental setups and do not provide standardized robotics datasets that combine camera, IMU, LiDAR, and geomagnetic measurements for reproducible research. **Across existing magnetic datasets and magnetic SLAM studies, three key limitations remain:**

- Lack of synchronized conventional robotics sensors (e.g., camera, LiDAR, and IMU) and tri-axis magnetometer measurements,
- Absence of structured repeated trajectories for drift and repeatability analysis for the magnetometer measurements in outdoor environments,
- No robotics-oriented dataset providing paired multi-modal sensing and accurate ground truth for magnetic-aided SLAM research.

Consequently, the role of geomagnetic sensing in mitigating yaw drift, improving loop closure, and supporting multi-modal state estimation remains insufficiently studied in the context of standard SLAM datasets.

To address this gap, we introduce the **Mag4D-SLAM Dataset**, a multi-modal geomagnetic dataset designed for magnetic-aware localization and mapping research. The dataset provides synchronized camera, LiDAR, IMU, and tri-axis magnetic measurements recorded along structured circular and rectangular trajectories with accurate ground-truth poses. The “4D” designation reflects the spatio-temporal nature of the dataset: the three-axis magnetic field $\mathbf{M}(t) = [m_x(t), m_y(t), m_z(t)]^T$ is recorded continuously along repeated traversals across multiple sessions, forming a long-term spatial magnetic field map. Long-term scene understanding and re-localization have been recognized as critical challenges in SLAM [5, 14, 15], yet no existing magnetic dataset provides the multi-session repeated-traversal structure necessary to study geomagnetic sensing as a long-term localization modality.

Fig. 1 illustrates the dataset concept and sensing paradigm. The Earth’s geomagnetic field provides a globally consistent yet spatially varying reference, observed in the robot frame as (M_x, M_y, M_z) . When recorded along structured trajectories, these measurements form location-dependent magnetic signatures. Despite variations in environmental conditions (e.g., day and night), the underlying magnetic field structure remains largely stable, producing repeatable environmental fingerprints that can help constrain heading drift and improve

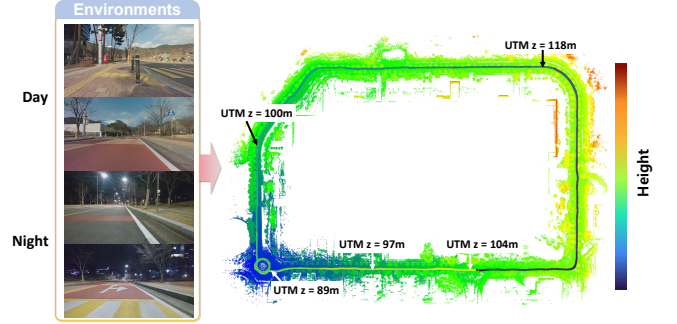


Fig. 2. Data collection environments and trajectory overview. Left: representative day and night camera images. Right: top-view LiDAR point cloud of the campus loop, colored by height, showing mild elevation changes across the trajectory.

cross-session consistency in multi-sensor SLAM.

The contributions of this work are summarized as follows:

- **First large-scale outdoor geomagnetic dataset:** Mag4D-SLAM provides 14 sequences totaling over 18 km of synchronized camera, LiDAR, IMU, tri-axis magnetometer, and GNSS data with accurate 6-DoF ground truth, collected on a quadruped robot platform across structured campus trajectories.
- **Repeated-traversal multi-condition design:** The dataset captures paired day and night sessions in both forward and reverse directions along identical paths with mild elevation changes across the trajectory, as shown in Fig. 2, enabling systematic analysis of geomagnetic consistency, heading-dependent signatures, and illumination-invariant localization.
- **Baseline analysis of geomagnetic cues:** We quantitatively evaluate three core properties relevant to magnetic-aware localization and mapping: 1) bounded global heading estimation without accumulated drift, 2) cross-session magnetic repeatability, and 3) location-discriminative place recognition. These baselines characterize how geomagnetic measurements can support future magnetic-aided multi-sensor fusion and SLAM research.

II. RELATED WORK

A. Benchmark Datasets for SLAM

Progress in SLAM has been strongly driven by standardized datasets providing synchronized multi-modal sensing

and reliable ground truth. KITTI [2] established a large-scale outdoor benchmark for visual and LiDAR-based SLAM using GPS/INS references. EuRoC [3] and TUM-VI [4] provide high-quality visual-inertial data, yet are limited in spatial scale and primarily indoor scenarios. Oxford Robot-Car [5] enables large-scale multi-season evaluation, and Newer College [8] offers centimeter-level ground truth via map alignment. More recent datasets such as Boreas [6] and TartanAir [16] further extend diversity across weather conditions and synthetic environments. Despite their contributions, none incorporate geomagnetic measurements as a core sensing modality, limiting their suitability for magnetic-assisted localization and heading estimation.

B. Magnetic Sensing for Localization and SLAM

Magnetic sensing has been explored as a complementary modality where GNSS is unreliable. Unlike vision or LiDAR, magnetometers capture spatial distortions of the Earth’s magnetic field caused by ferromagnetic structures, producing location-dependent signatures that support localization, loop closure detection, and global heading correction. More recent research has incorporated magnetometers into inertial navigation systems and probabilistic mapping frameworks for SLAM [17], demonstrating that magnetic anomalies can serve as spatial features for pose correction. Tan *et al.* [18] propose a magnetometer-based approach for high-angular-rate estimation during gyroscope saturation, though their work addresses dynamic attitude recovery rather than spatial magnetic field analysis. Magnetometers have also been used to mitigate yaw drift in visual-inertial pipelines [19]. While these works demonstrate the utility of magnetic sensing in SLAM, they rely on task-specific setups and lack publicly available multi-modal datasets for reproducible research.

C. Magnetic Datasets

Several collections have been proposed to support magnetic-based localization, yet none fully address the requirements of robotic SLAM research. UJIIndoorLoc-Mag [10] introduced one of the first public magnetic fingerprinting databases, containing magnetometer and IMU measurements along indoor corridors. MagPIE [7] extended this by providing centimeter-level ground truth across multiple buildings, though it remains focused on indoor localization without synchronized LiDAR-IMU-magnetometer measurements. MagWi [9] introduced a long-term magnetic and Wi-Fi dataset across heterogeneous smartphones and multi-floor buildings, targeting temporal stability rather than robotic SLAM research. More recently, Mag-Match [11] extracts orientation-invariant features from 3D magnetic vector fields via a physics-informed Gaussian Process for map matching, but does not provide a large-scale outdoor dataset for magnetic-aware SLAM. As summarized in Table I, Mag4D-SLAM is designed to address these gaps by providing large-scale outdoor trajectories, repeated traversals under day and night conditions, and globally aligned 6-DoF ground-truth poses.

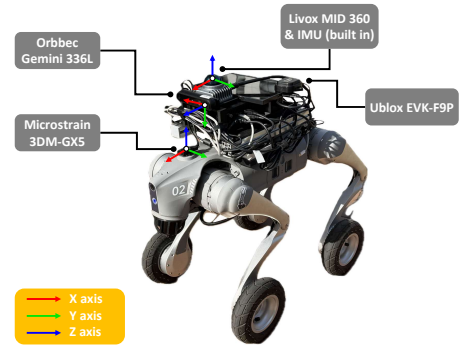


Fig. 3. Robot platform with all sensors: LiDAR with built-in IMU, GNSS receiver, camera, and magnetometer.

TABLE II. Overview of the sensor suite used in our system

Sensor	Type	Rate	Characteristics
Camera	Gemini 336L	30 Hz	RGB 1280×800, H94°/V68°
LiDAR	Mid-360	10 Hz	360°/59° FOV; 200k pts/s; 70 m range
IMU	Mid-360 (built-in)	200 Hz	3-axis Gyroscope; 3-axis Accelerometer
Magnetometer	3DM-GX5-AHRS	100 Hz	3-axis, Temp. compensated ±8 Gauss;
GNSS	EVK-F9P	1 Hz	Multi-band GNSS receiver; Operated in standalone mode (~meter-level accuracy)
SE(3) Ground-truth Pose	LiDAR-inertial localization on HD map	10 Hz	Multi-session map-based verification by human

III. SYSTEM OVERVIEW

A. Sensor Suite

As shown in Fig. 3, the Mag4D platform consists of a rigidly mounted multi-modal sensor payload including a 3D LiDAR, a monocular/stereo camera, a 6-axis IMU (3-axis accelerometer, 3-axis gyroscope), a tri-axis magnetometer, and a GNSS receiver. Table II summarizes the multi-sensor configuration used in our platform with LiDAR-inertial ground-truth pose estimation. All sensors are rigidly attached to a common base frame and recorded into a single ROS 2 bag file.

B. Ground Truth Pose Generation

Following the approach of the Newer College dataset [8], we generate precise 6-DoF ground-truth poses by registering mobile LiDAR scans against a high-definition (HD) prior map. The HD prior map is constructed from mapping-vehicle LiDAR scans using RTK-GPS fixed-solution trajectory constraints and pose-graph optimization. Specifically, each mobile LiDAR scan is aligned to the prior map using ICP:

$$T^* = \arg \min_{T \in SE(3)} \sum_i \| \mathbf{p}_i^{map} - T \mathbf{p}_i^{scan} \|^2. \quad (1)$$

To ensure path smoothness between registered poses, we used RKO-LIO [20] as the localizer, providing continuous LiDAR-inertial estimates that were subsequently refined via scan-to-map ICP alignment. Registration accuracy shows

TABLE III. Sequence-level statistics of our dataset, covering three environments, two lighting conditions, and three traversal directions.

Seq.	Env.	Light.	Dir.	Path (m)	Duration (s)
campus_day_fwd_01	Campus	☀ Day	→ Fwd	1382.9	1203.8
campus_day_fwd_04	Campus	☀ Day	→ Fwd	1554.06	1291.9
campus_day_fwd_08	Campus	☀ Day	→ Fwd	1465.62	1181.2
campus_day_fwd_09	Campus	☀ Day	→ Fwd	1443.59	1132.3
campus_day_rev_05	Campus	☀ Day	← Rev	1637.24	1299.1
campus_day_rev_07	Campus	☀ Day	← Rev	1605.33	1234.9
campus_night_fwd_00	Campus	☾ Night	→ Fwd	1376.6	1117.3
campus_night_rev_11	Campus	☾ Night	← Rev	1527.28	1224.4
circle_day_fwd_rev_02	Circle	☀ Day	↔ Fwd+Rev	1119.1	822.1
circle_day_fwd_rev_06	Circle	☀ Day	↔ Fwd+Rev	987.53	763.0
circle_day_fwd_rev_10	Circle	☀ Day	↔ Fwd+Rev	1402.79	993.2
circle_night_fwd_rev_00	Circle	☾ Night	↔ Fwd+Rev	1064.4	816.9
library_day_fwd_03	Library	☀ Day	→ Fwd	189.52	163.3
campus_circle_night_rev_12	Campus+Circle	☾ Night	← Rev	1676.00	1406.2

*circle_night_fwd_rev_00 and campus_night_fwd_00 were intentionally collected without magnetometer calibration to illustrate the effect of hard-iron and soft-iron distortions, as visualized in Fig. 4.

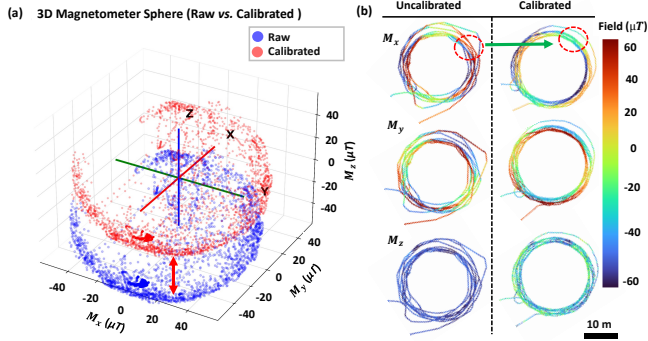


Fig. 4. (a) 3D visualization of uncalibrated (raw) (blue) and calibrated (red) magnetic measurements. Calibration removes hard- and soft-iron distortions, transforming the ellipsoidal distribution into a centered sphere. (b) 2D projections of M_x , M_y , and M_z , before and after calibration, demonstrating improved symmetry and heading consistency after distortion removal.

a standard deviation below 5cm, consistent with high-precision benchmarks [8]. The ground-truth quality was further verified by overlaying each trajectory-based session map against the HD prior map in CloudCompare, with manual inspection confirming spatial alignment across all sequences.

IV. SENSOR CALIBRATION

Multi-modal extrinsic calibrations are essential for reliable sensor fusion and SLAM evaluation. We perform intrinsic, extrinsic, and temporal calibration following established dataset practices [4, 15]. Specifically, camera-IMU and LiDAR-IMU extrinsic transformations are estimated using motion-based hand-eye calibration. Magnetometer calibration is additionally performed as described below, given its critical role in heading estimation.

Magnetometer calibration background: Raw magnetometer readings are affected by hard-iron bias, soft-iron deformation, axis misalignment, and scale errors [19, 21], which

introduce systematic yaw errors over long trajectories without correction. The measured magnetic vector is modeled as:

$$\mathbf{m}_{meas} = \mathbf{S}\mathbf{m}_{true} + \mathbf{b} + \mathbf{n} \quad (2)$$

where $\mathbf{b} \in \mathbb{R}^3$ is the hard-iron bias, $\mathbf{S} \in \mathbb{R}^{3 \times 3}$ is the distortion matrix encoding soft-iron effects, axis misalignment, and scale errors, and $\mathbf{n}$ is measurement noise. The calibrated measurement is obtained via ellipsoid fitting [22]:

$$\mathbf{m}_{cal} = \mathbf{S}^{-1}(\mathbf{m}_{meas} - \mathbf{b}), \quad \|\mathbf{m}_{cal}\| \approx B \quad (3)$$

where B is the local geomagnetic magnitude.

Magnetometer calibration procedure: Calibration was performed in-situ with all onboard electronics active to capture realistic platform-induced distortions from motors, batteries, and metallic structures [23]. The robot was manually rotated through full yaw, pitch, and roll motions to ensure sufficient excitation of all three magnetic axes for reliable ellipsoid fitting.

Calibration analysis: Fig. 4(a) shows the 3D magnetic measurement distribution before and after calibration. The uncalibrated measurements form a distorted ellipsoidal distribution offset from the origin, indicating hard-iron bias and soft-iron deformation. After correction, the measurements approximate a centered sphere, satisfying $\|\mathbf{m}_{cal}\| \approx B$. Fig. 4(b) shows the spatial magnetic component maps (M_x , M_y , M_z) before and after calibration along a circular trajectory. The calibrated maps exhibit improved symmetry and heading consistency, confirming that ellipsoid fitting effectively compensates for platform-induced distortions.

V. MAG4D DATASET

Environment and Trajectory Design: Data collection was conducted in a large structured outdoor environment covering the university campus road segments and pedestrian areas. The primary trajectory includes a circular road segment surrounding a central roundabout and rectangular sections with mild elevation changes, providing realistic terrain conditions within a controlled geometric layout. Shorter sequences

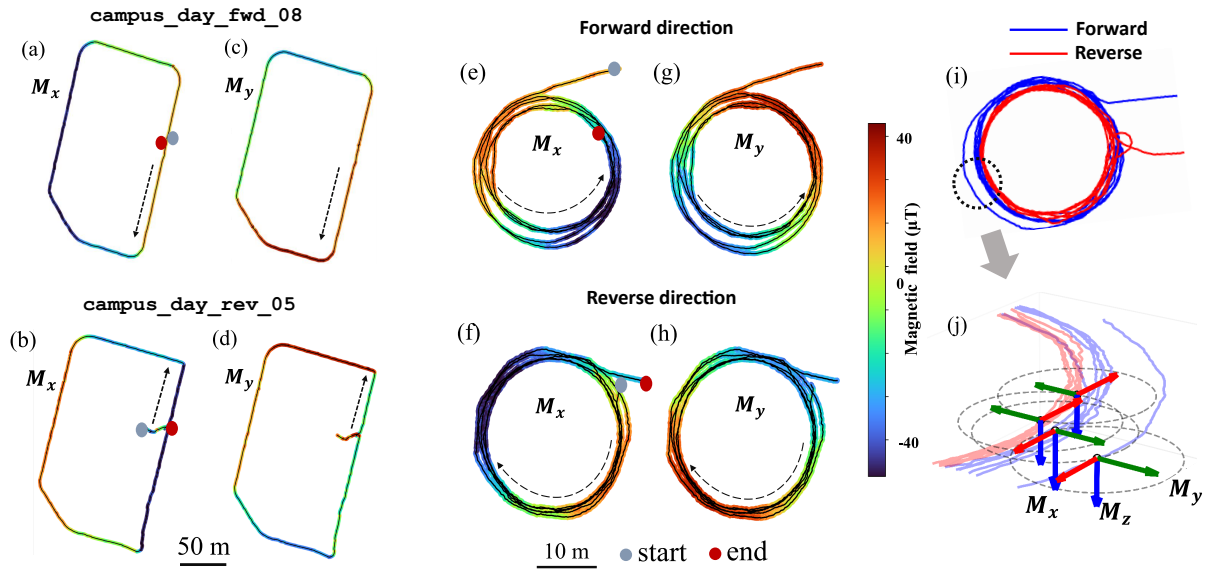


Fig. 5. Magnetic field component maps along forward and reverse traversals in the Mag4D-SLAM dataset. (a–d) Rectangular campus trajectory showing the spatial distribution of M_x and M_y during forward and reverse runs. (e–h) Circular trajectory executed in forward and reverse directions, illustrating the corresponding magnetic field component maps. Color represents the magnetic field intensity (μT), while dashed arrows indicate the direction of motion. Start and end positions are indicated by gray and red markers. (i) Trajectory of the forward (blue) and reverse (red) runs near the circle, highlighting traversal direction. (j) Zoom-in region showing the magnetic field vector at a selected location, where the three-axis components M_x , M_y , and M_z are visualized using red, green, and blue arrows, respectively.

recorded near the campus library offer localized high-overlap measurements for fine-grained spatial consistency analysis.

The dataset includes 14 sequences across repeated traversals of identical paths under both daytime and nighttime conditions, with forward and reverse directions as summarized in Table III. While visual sensing performance varies across lighting conditions, magnetic field measurements remain continuously available, enabling systematic evaluation of illumination-invariant localization.

Data Format and Availability: The Mag4D-SLAM dataset is provided in both raw ROS bag format and processed CSV format (e.g., `mag.csv`, `gt_pose.csv`). Each sequence contains synchronized multi-sensor recordings and associated ground-truth files. The dataset is publicly available.¹

VI. EXPERIMENTAL ANALYSIS

This section quantitatively validates three core properties of the Mag4D-SLAM dataset: multi-condition magnetic repeatability, drift-free heading estimation, and place recognition capability.

A. Multi-Condition Repeatability Analysis

We evaluate magnetic field variability under forward and reverse traversals for rectangular (campus) trajectories on `campus_day_fwd_08` and `campus_day_rev_05`. Fig. 5(a–d) presents the magnetic field component maps for M_x and M_y along a rectangular campus loop executed in forward and reverse directions. The color scale represents magnetic field intensity (μT) at each spatial location. Although the geometric trajectory is identical, the measured

components depend strongly on the robot’s heading, which provides an informative directional signature. This property is particularly valuable when visual SLAM performance degrades under low-light or nighttime conditions.

For the circular trajectory (Fig. 5(e–h)), clockwise (reverse) and counter-clockwise (forward) motions from the `circle_day_fwd_rev_10` produce distinct magnetic component distributions despite overlapping spatial paths. Specifically, M_x and M_y exhibit sign changes and intensity redistribution along identical spatial segments, with variations spanning approximately $\pm 40 \mu\text{T}$. These differences arise from changes in body-frame orientation rather than environmental inconsistency. Notably, within a single direction of motion, the magnetic intensity at specific spatial locations remains consistent, demonstrating the uniqueness and repeatability of the magnetic signature. The zoomed visualization in Fig. 5(j) further illustrates this behavior: at nearly identical spatial positions, the magnetic vectors (red and green arrows) differ between traversals, with M_x and M_y appearing approximately inverted. This confirms that magnetometer measurements are frame-dependent and strongly influenced by heading, even when the robot revisits the same physical location. Similar orientation-dependent magnetic signatures have also been reported in magnetic field based loop-closure studies [24].

Importantly, despite component-level differences, the overall spatial structure of the magnetic field remains continuous along the trajectory, while heading-dependent differences are particularly pronounced between forward and reverse traversals. The absence of abrupt spatial discontinuities indicates that the environmental magnetic field varies smoothly across the surveyed area. As a result, while the measured com-

¹<https://mag4d-dataset.github.io/Mag4d-Dataset/>

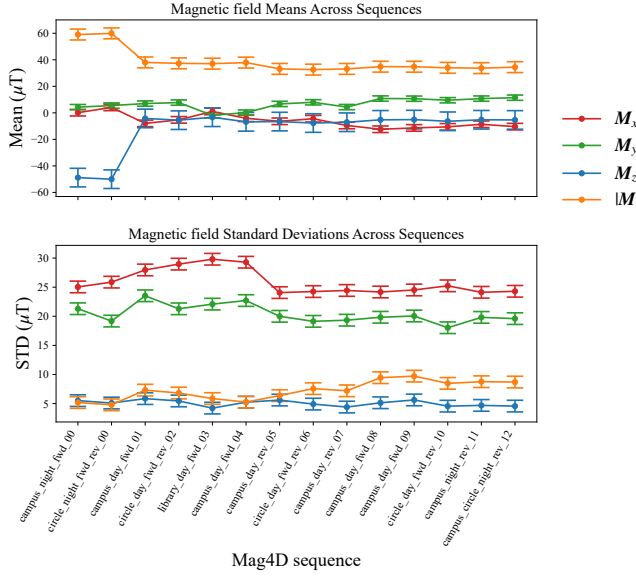


Fig. 6. Magnetic field statistics across all Mag4D-SLAM dataset sequences. **Top:** Mean values of the magnetic components (M_x , M_y , M_z) and the magnetic field magnitude ($|M|$) for each sequence, with error bars indicating the standard error. **Bottom:** Standard deviation of the magnetic components and the field magnitude across the same sequences, illustrating the variability of the geomagnetic measurements along different trajectories and environmental conditions (day/night and forward/reverse traversals).

ponents change with robot heading due to projection onto the sensor frame, the underlying magnetic field distribution remains spatially consistent.

These results demonstrate that magnetic measurements are direction-sensitive at the component level yet spatially repeatable in structure, an essential property for magnetic-aware SLAM and heading-consistent loop closure.

B. Quantitative Magnetic Stability Analysis

Fig. 6 summarizes the mean and standard deviation of M_x , M_y , M_z , and $|M|$ across all sequences.

The first two sequences `campus_night_fwd_00` and `circle_night_fwd_rev_00`, recorded without calibration, exhibit clear offset distortion. In particular, M_z shows a strong negative bias of $\approx -50 \mu\text{T}$, while $|M|$ exceeds $60 \mu\text{T}$, indicating hard-iron and scale effects. The larger spread of $|M|$ confirms violation of the constant-magnitude geomagnetic constraint.

After calibration, the mean components stabilize: M_x and M_z remain within $\pm 15 \mu\text{T}$, while M_y stays within $\pm 10 \mu\text{T}$. The magnitude $|M|$ consistently clusters between 35 and $45 \mu\text{T}$, matching the expected local field strength.

The standard deviation remains consistent across the calibrated runs. σ_{M_x} ranges from 24–30 μT , σ_{M_y} from 18–23 μT , and σ_{M_z} remains lowest at 4–7 μT , reflecting its vertical alignment with the Earth’s field. The magnitude variation is tightly bounded ($\sigma_{|M|} \approx 5\text{--}10 \mu\text{T}$), indicating improved spherical consistency. Overall, calibration enforces $\|\mathbf{m}_{\text{cal}}\| \approx B$ (local geomagnetic field), reduces bias spread, and stabilizes magnetic intensity, supporting repeatable signatures for magnetic-aided SLAM.

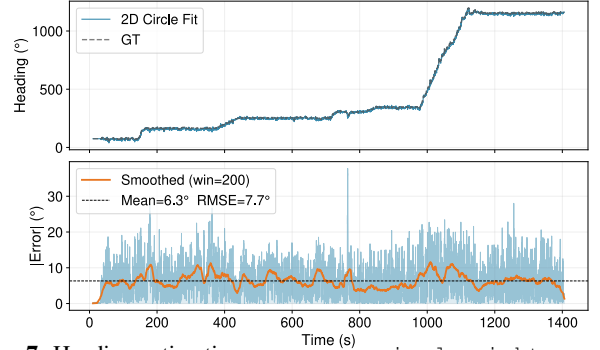


Fig. 7. Heading estimation on `campus_circle_night_rev_12` using the proposed online 2D circle fitting calibration. **Top:** estimated heading tracks ground truth (GT) continuously without drift over the full 1400 s trajectory. **Bottom:** absolute heading error with a moving average filter (window = 200 samples) overlaid; mean error 6.3° and RMSE 7.7° remain bounded throughout, attributable to local magnetic anomalies along the campus route.

TABLE IV. Per-Segment RMSE Across Representative Sequences

Sequence Segment	campus_day_rev_05	campus_day_fwd_08	campus_circle_night_rev_12
0–20%	4.8°	4.6°	7.1°
20–40%	5.3°	6.2°	8.4°
40–60%	5.6°	6.7°	7.3°
60–80%	4.9°	7.0°	8.3°
80–100%	5.9°	5.7°	7.0°

TABLE V. Heading Estimation Accuracy (RMSE)

Method	Day (Seq. 05)	Day (Seq. 08)	Night (Seq. 12)
Direct + Hard-iron	6.5°	6.3°	12.0°
Direct + 2D Circle	5.3°	6.1°	7.7°

Seq. 05: `campus_day_rev_05`.

Seq. 08: `campus_day_fwd_08`.

Seq. 12: `campus_circle_night_rev_12`.

C. Drift-Free Heading Estimation

A key motivation for Mag4D is to enable global heading estimation without yaw drift. We quantitatively evaluate this property by comparing two heading estimation strategies against the ground-truth (GT) yaw described in Sec. III-B.

1) **Methods: Direct + Hard-Iron Correction.** We apply tilt-compensated [25] heading estimation directly from calibrated magnetic measurements without sequential filtering. The hard-iron bias is estimated per-sequence using the min/max method:

$$b_i = \frac{\max(M_i) + \min(M_i)}{2}, \quad i \in \{x, y, z\}. \quad (4)$$

The tilt-compensated heading is then computed as:

$$\begin{aligned} M_x^h &= M_x \cos \theta + M_y \sin \phi \sin \theta + M_z \cos \phi \sin \theta \\ M_y^h &= M_y \cos \phi - M_z \sin \phi \\ \hat{\psi} &= \text{atan2}(-M_y^h, M_x^h), \end{aligned} \quad (5)$$

where ϕ and θ denote roll and pitch from the IMU.

Direct + Online 2D Circle Fitting. Ground robots exhibit small roll/pitch variation ($|\phi| < 5^\circ$, $|\theta| < 13^\circ$) but full 360° yaw coverage during normal traversal, such that the M_x – M_y

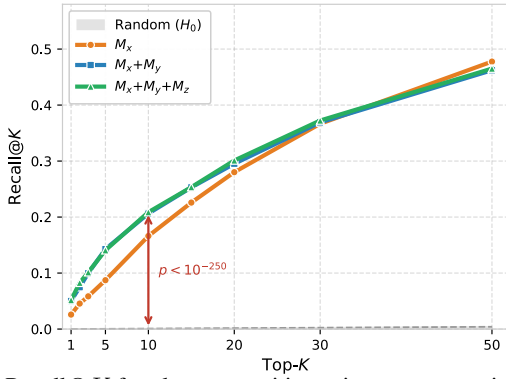


Fig. 8. Recall@ K for place recognition using raw magnetic features (campus_day_rev_05 vs. campus_night_rev_11, day-night cross-session). Adding axes progressively improves retrieval performance, with $M_x + M_y + M_z$ achieving Recall@10=20.9%, demonstrating that raw magnetic signatures carry location-discriminative information as a baseline for learning-based methods.

plane is naturally well-sampled. We exploit this property to fit a 2D circle online [26]:

$$d_k = \sqrt{(M_x^k - c_x)^2 + (M_y^k - c_y)^2}$$

$$(c_x^*, c_y^*, r^*) = \arg \min_{c_x, c_y, r} \sum_k (d_k - r)^2 \quad (6)$$

yielding a calibrated center (c_x^*, c_y^*) subtracted before applying Eq. (5). Crucially, this requires *no dedicated calibration maneuver*—normal operation provides sufficient excitation.

2) *Drift-Free Property:* Fig. 7 illustrates the estimated yaw against GT on campus_circle_night_rev_12. Table V summarizes RMSE across two sequences. The online 2D circle fitting consistently achieves lower RMSE, with a particularly pronounced improvement on campus_circle_night_rev_12 (7.7° vs. 12.0°).

Per-segment RMSE further confirms the drift-free nature of the proposed approach (Table IV): error remains consistently bounded across all segments regardless of traversal duration.

3) *Discussion:* The results reveal two key findings. First, online 2D circle-fitting calibration consistently outperforms the standard hard-iron method, particularly on longer sequences, demonstrating that *calibration quality is the dominant factor* in magnetic heading accuracy. Second, ground robots naturally generate sufficient magnetic excitation during normal traversal to enable online calibration, eliminating the need for dedicated calibration maneuvers. The residual error of $\sim 6^\circ$ – 8° is attributable to local magnetic anomalies from campus structures. This analysis confirms that Mag4D-SLAM dataset provides the structured trajectories and high-precision ground truth necessary to quantitatively evaluate global heading observability—a capability absent from existing SLAM benchmarks.

D. Place Recognition Baseline

To assess whether raw magnetic signatures carry location-discriminative information, we formulate a null hypothesis H_0 : retrieval performance is equivalent to random selection from the database. We evaluate cross-

session place recognition between campus_day_rev_05 and campus_night_rev_11 using nearest-neighbor retrieval with Euclidean distance in magnetic feature space. A query pose is considered correctly retrieved if at least one of the top- K candidates lies within 3 m of the ground-truth position. As shown in Fig. 8, all three feature configurations substantially exceed the random baseline ($p_0 = K/|\mathcal{D}|$, $|\mathcal{D}| = 12,245$) across all K . At $K = 10$, $M_x + M_y + M_z$ achieves Recall@10=20.9% versus a random baseline of 0.08%, a $256\times$ improvement. A one-sided binomial test yields $p < 10^{-250}$ for all tested K , strongly rejecting H_0 and confirming that raw magnetic measurements are statistically significant location descriptors. Notably, M_x alone achieves competitive Recall@50 of 47.8% compared to the full three-axis feature (46.5%), suggesting that M_x carries the most location-discriminative information among the three components. These results establish a raw-feature baseline and motivate the development of learning-based magnetic descriptors using the Mag4D benchmark.

VII. DISCUSSION

The experimental results demonstrate that geomagnetic sensing provides bounded heading estimates without accumulated drift and location-discriminative signatures across diverse operational conditions. We outline three directions for integrating magnetic sensing into robotic SLAM pipelines.

Global Heading Correction. As demonstrated in Sec. VI-C, magnetometer-derived heading achieves RMSE of 6–8° without accumulating drift over trajectories exceeding 1.4 km, offering a level of long-term consistency that IMU-based orientation cannot provide. This property makes magnetic sensing a practical absolute orientation reference for visual-inertial and LiDAR-inertial odometry systems operating in GNSS-denied environments.

Magnetic Place Recognition. Raw magnetic signatures achieve Recall@10=20.9% in cross-session place recognition, providing a viable baseline for magnetic-aided loop closure. In conditions where visual and LiDAR descriptors degrade—including low-light operation, structurally repetitive environments, and adverse weather—magnetic fingerprints offer an infrastructure-free retrieval signal independent of ambient light levels. Learning-based descriptors trained on the Mag4D benchmark hold promise for substantially improving upon this baseline.

Long-Term Localization. Geomagnetic signatures exhibit temporal stability that visual maps often struggle to maintain under visual degradation, seasonal variation, or structural modifications. The cross-session consistency observed in our experiments suggests that a magnetic map constructed from a single traversal can support reliable re-localization across multiple sessions, making magnetic sensing a robust modality for long-term autonomous operation. Nevertheless, environments with strong ferromagnetic interference or dense metallic infrastructure may reduce signature reliability, a limitation that warrants further investigation.

VIII. CONCLUSION

This paper presented the **Mag4D-SLAM** dataset, a large-scale outdoor magnetic-aware SLAM dataset for studying magnetic-assisted localization. Unlike existing magnetic datasets limited to small indoor environments, Mag4D-SLAM provides structured outdoor trajectories with accurate 6-DoF ground truth collected across a university campus under both day and night conditions. The dataset includes synchronized LiDAR, IMU, and tri-axis magnetometer measurements, enabling evaluation of magnetic-assisted heading correction, place recognition, and drift mitigation in modern SLAM systems. Our analysis demonstrates that magnetic signatures remain spatially consistent across repeated traversals and can complement visual-inertial and LiDAR-inertial pipelines for global yaw stabilization. By releasing the Mag4D-SLAM dataset, we aim to support future research on magnetic-aware SLAM and encourage the use of geomagnetic sensing as a robust, infrastructure-free modality for long-term robotic autonomy. Although the current dataset includes repeated day/night and directional traversals, it does not yet cover long-term seasonal variations, weather changes, or large temporal gaps across months. Therefore, the present study primarily focuses on evaluating cross-session magnetic consistency under varying illumination and traversal conditions within a structured outdoor campus environment.

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